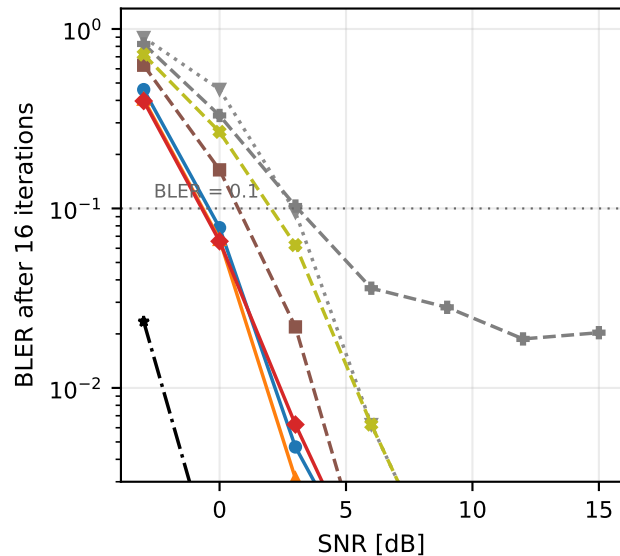
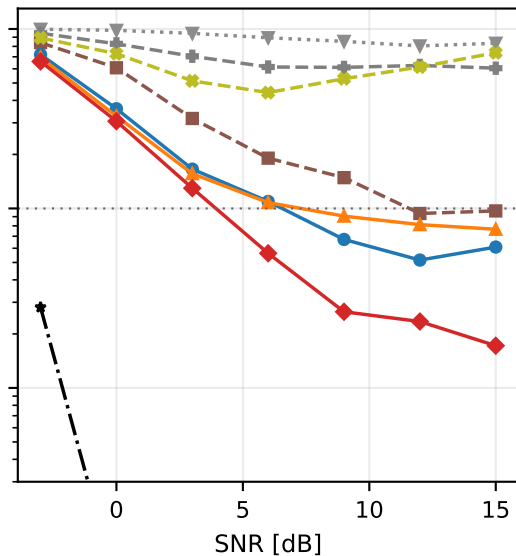


D1: sparse specular multipath,  $8 \times 4$  MIMO, QPSK, rate-1/2  $(133, 171)_8$ ,  $T = 16$ , 16 iterations

(a) headline cell:  $T_p = 4$



(b)  $T_p = 2$ : first pass uninformative



- ▽·· R0 pilot-only LMMSE, 1 pass (no turbo loop)
- R1 classical turbo (APP feedback, no LOO)
- +— R3 BiG-AMP + BCJR (i.i.d. prior, structural)
- \*— R4 3-module SC-VAMP-type (Onsager interface)
- R2 D-15+LOO, Gaussian sample-cov. prior
- ▲— M Module H = GMM, K=32 (literal)
- ◆— M Module H = GMM,  $b^*$  (val. log-lik.)
- ★— R5 genie CSI (lower bound)